

1. Let  $(V, + \cdot)$  be a vector space over the field  $\mathbb{K} := \mathbb{R}$  or  $\mathbb{C}$ . Define what is meant by the statement that the mapping  $N : V \rightarrow \mathbb{R}$  is a norm on  $V$ .
2. Let  $(V, \|\cdot\|)$  be a normed space over the field  $\mathbb{K} = \mathbb{R}$  or  $\mathbb{C}$ . Set

$$d(u, v) := \sqrt{\|u - v\|} \quad \forall u, v \in V.$$

- (a) Show that  $d$  is a distance on  $V$ . (Hint:  $\sqrt{s+t} \leq \sqrt{s} + \sqrt{t}$  for every  $s, t \geq 0$ .)
- (b) Is the distance  $d$  induced by a norm on  $V$ ? Explain clearly your answer.

3. Let  $N : \mathbb{R}^2 \rightarrow \mathbb{R}$  be defined by  $N(x, y) := \int_0^1 |x + ty| \, dt \quad \forall (x, y) \in \mathbb{R}^2$ .  
Show that  $N$  is a norm on  $\mathbb{R}^2$ .

4. Let  $V = \mathcal{C}([0, 1], \mathbb{R})$ , the space of continuous functions from  $[0, 1]$  to  $\mathbb{R}$ . Set

$$\|f\|_1 := \int_0^1 |f(t)| \, dt \quad \text{and} \quad \|f\|_\infty = \sup_{0 \leq t \leq 1} |f(t)| \quad \forall f \in V.$$

- (a) Show that  $\|\cdot\|_1$  and  $\|\cdot\|_\infty$  are norms on  $V$ .
- (b) Is the mapping  $N(f) := \left| \int_0^1 f(t) \, dt \right|$  also a norm on  $V$ ? Explain your answer.
- (c) Let  $(f_n)$  be the sequence in  $V$  defined by  $f_n(t) := t^n$  for any  $t \in [0, 1]$ .
  - (i) Find the pointwise limit of  $(f_n)$ .
  - (ii) Does the sequence  $(f_n)$  converge in  $(V, \|\cdot\|_\infty)$ ? Explain clearly your answer.
  - (iii) Does the sequence  $(f_n)$  converge in  $(V, \|\cdot\|_1)$ ? Explain clearly your answer. (Hint: Find  $\|f_n\|_1$ ).
5. (a) Define what is meant by the statement that the normed space  $(V, \|\cdot\|)$  is a Banach space.
- (b) Let  $(V, \|\cdot\|_V)$  and  $(W, \|\cdot\|_W)$  be two Banach spaces over the same field  $\mathbb{K} = \mathbb{R}$  or  $\mathbb{C}$ . Consider the mapping  $N : V \times W \rightarrow \mathbb{R}$  defined by

$$N(v, w) := \|v\|_V + \|w\|_W \quad \forall (v, w) \in V \times W.$$

- (i) Show that  $N$  is a norm on  $V \times W$ .

(ii) Show  $V \times W$  is a Banach space when equipped with the norm  $N$ .

6. Let

$$\ell^\infty(\mathbb{R}) := \{x = (x_n) \text{ sequence of real numbers such that } \sup_{n \in \mathbb{N}} |x_n| < +\infty\},$$

$$\ell^p(\mathbb{R}) := \left\{ x = (x_n) \text{ sequence of real numbers such that } \sum_{n=0}^{\infty} |x_n|^p < +\infty \right\} \text{ with } 1 \leq p < \infty.$$

(a) Say which of the following sequences belong(s) to  $\ell^1(\mathbb{R})$ :

$$(i) \ x_n = \frac{2^n}{n + 9^{n/3}} \quad \forall n \in \mathbb{N}; \quad (ii) \ \begin{cases} y_0 = 0 \\ y_n := \frac{1}{n} \arctan\left(\frac{1}{\sqrt{n}}\right) \end{cases} \quad \forall n \in \mathbb{N} \setminus \{0\}.$$

(Hint:  $\arctan x \leq x \quad \forall x \geq 0$ ).

(b) Show that  $\ell^p(\mathbb{R}) \subset \ell^q(\mathbb{R})$  for  $1 \leq p \leq q \leq \infty$ .

7. We recall that

$$\|(x, y)\|_\infty := \max(|x|, |y|) \quad \text{and} \quad \|(x, y)\|_1 := |x| + |y| \quad \forall (x, y) \in \mathbb{R}^2.$$

For  $n \in \mathbb{N}$ , we set

$$u_n := \left( \frac{1}{1+n}, 1 + e^{-n} \right) \in \mathbb{R}^2.$$

(a) Study the convergence of the sequence  $(u_n)$  in  $\mathbb{R}^2$  equipped with the norm  $\|\cdot\|_\infty$ .

(b) Study the convergence of the sequence  $(u_n)$  in  $\mathbb{R}^2$  equipped with the norm  $\|\cdot\|_1$ .